

$$a) \leq \sqrt{\left\{ \left[\sum_{i=1}^n [(x_i - y_i)^2] + [(y_i - z_i)^2] \right\}}$$

(by rearranging terms, we get)

$$\leq \sqrt{\left\{ \left[\sum_{i=1}^n [(x_i - y_i)^2] + \left[\sum_{i=1}^n [(y_i - z_i)^2] \right] \right\}}$$

$$= d_E(x, y) + d_E(y, z)$$

Hence, the proof.

3)

$$S(a, 1) = \{(x = (x_1, x_2), \text{ with } |x_1 - a_1| + |x_2 - a_2| < 1\}$$

$$S^c(a, 1) = \{(x = (x_1, x_2), \text{ with } |x_1 - a_1| + |x_2 - a_2| \leq 1\}$$

4) **Every open sphere $S(a, r)$ is an open set:**

By definition, we know that for $S(a, r)$ to be an **open set**, we need to show that for each element x of $S(a, r)$, there is an open interval $]x - \varepsilon, x + \varepsilon[\subseteq S(a, r)$, with $\varepsilon > 0$.

Let $x \in S(a, r)$, then, $d(a, x) < r$.

if $x = a$, then $S(x, r) = S(a, r) \subseteq S(a, r)$.

if $x \neq a$, then $d(x, a) > 0$, and $d(a, x) < r$, (from above).

Let $\varepsilon = d(x, a) > 0$, then, $\varepsilon < r$, or $r - \varepsilon > 0$.

we show that $]x - (r - \varepsilon), x + (r - \varepsilon)[\subseteq S(a, r)$.

Let $y \in]x - (r - \varepsilon), x + (r - \varepsilon)[$,

then $d(x, y) \leq r - \varepsilon$.

and $d(a, y) \leq d(a, x) + d(x, y) \leq \varepsilon + (r - \varepsilon) = r$

implies $y \in S(a, r)$

Thus, $y \in]x - (r - \varepsilon), x + (r - \varepsilon)[$ implies $y \in S(a, r)$

Or, $]x - (r - \varepsilon), x + (r - \varepsilon)[\subseteq S(a, r)$, for every $x \in S(a, r)$.

Hence, the proof.